Evolution and Mutation

30A



Genes are the potential of an organism. The environment may or may not allow the full development of this potential. Genes are contained in chromosomes in the nucleus of cells. Genes are short lengths of chromosomes and are the basic unit of inheritance. They usually copy themselves exactly each time a cell divides (meiosis and mitosis) but, occasionally, the unexpected happens — and the unexpected is usually (but not always) unpleasant.

Genes are made of deoxyribonucleic acid (DNA), which has a very complicated structure. If this structure is altered in any way, a different gene may result. This is called *mutation*. Alleles are different genes that influence one characteristic. Mutations can remove one allele and replace it with another.

Mutations very rarely occur spontaneously — perhaps one in a million may happen without apparent cause. The rate is increased by substances called *mutagens*. Some of these are produced by the cell themselves and others come from the environment. These include:

- some chemicals
- radiation
- increased temperature.

Chemicals that are suspected of being mutagens include nitrous acid (HNO_2) and nitroso compounds which are present in treated meats such as salami and bacon, mustard gas (used in warfare), tobacco tars, and fungi that spoil foods. (Some of these substances may also cause cancer — they are *carcinogenic*.)

Radiation includes gamma radiation (a form of nuclear radiation), X-rays, cosmic rays (from space) and ultraviolet (UV) light. There are also natural radioactive sources from the Earth itself. The ozone layer protects us from much of the harmful radiation from space, but this has been damaged by such man-made chemicals as CFCs (chlorofluorocarbons).

An increase in temperature can cause a five-fold increase in the rate of mutations. Violent temperature changes can cause an even greater change in rate.

If mutations occur in gametes (germ or sex cells), they are inherited. They are usually recessive and may initially go undetected. They will occur in the phenotype (appearance of the individual) if the homozygous condition (containing identical alleles) occurs. The possibility of this is increased by in-breeding.

The overwhelming majority of mutations are harmful to the organisms affected, sometimes (very rarely), however, they may be beneficial (helpful).

Natural selection

This has sometimes been referred to as 'survival of the fittest'. It is a process by which the environment selects favourable variations and eliminates (gets rid of) harmful or less successful ones. (Environment means the surroundings or conditions in which the organism lives.)

After many generations of the selection and 'weeding-out' of the less fit, the characteristics of a population may change and the population may become better adapted to the current environment. If an organism is suited to its environment and there are no external pressures, the organism can remain unchanged.

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Evolution and Mutation

30B

Charles Darwin was a scientist who, in the middle of the 19th century, after looking at animals in South America, the Galapagos Islands and Australia, came up with a theory of evolution to account for the diversity of animal life throughout the world. His theory can be summarised by the points below.

- In a population there are variations. All the members of a particular species are not indentical.
- In any generation, there are offspring that don't reach maturity (grow up) and reproduce. The characteristics of these organisms are removed from the population.
- The organisms that survive and reproduce are well adapted to that particular environment. They have favourable variations.
- Favourable variations are passed on to offspring. They become more and more common in the population.

Alfred Russel Wallace (1823–1913) was a British naturalist who also developed a theory of evolution based on natural selection, at the same time as Darwin. While working in the East Indies, Wallace noted some fundamental differences between Asian animal species and Australian species. He placed a hypothetical 'dividing line' known as *Wallace's Line* between the islands of Lombok and Bali (Indonesia), between Sulawesi and Borneo and between the Moluccas and the Philippines. Many mammals, fish and birds occur in large numbers on one side of Wallace's line but, a few kilometres on the other side of the line, they either don't occur or are in much fewer numbers.

1. Name the two processes involved in cell division and replication.
2. What is an 'allele'?
3. What does a 'mutagen' do? Give three specific examples of mutagens.
4. A 'carcinogen' causes what disease?
5. Give a definition of 'gamete'.
6. Give a definition of 'phenotype'.
7. Give a definition of 'mature'.
8. Are all mutations harmful? Explain your answer.
9. Summarise the four points in Darwin's 'Theory of Evolution'.
10. Why might animals like giraffes develop from relatively short-necked animals over a period of time when grazing is in short supply?
11. The crocodile has remained almost unchanged over 100 million years. Why might this be so?
12. What is the source of '*cosmic*' rays?
13. What does this statement define? '*A permanent change in hereditary material that involves either a physical change in chromosome relations or a biochemical change in genes.*'

Research

14. Photocopy or draw a map showing Australia and neighbouring Asian neighbours. Draw in Wallace's Line.

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